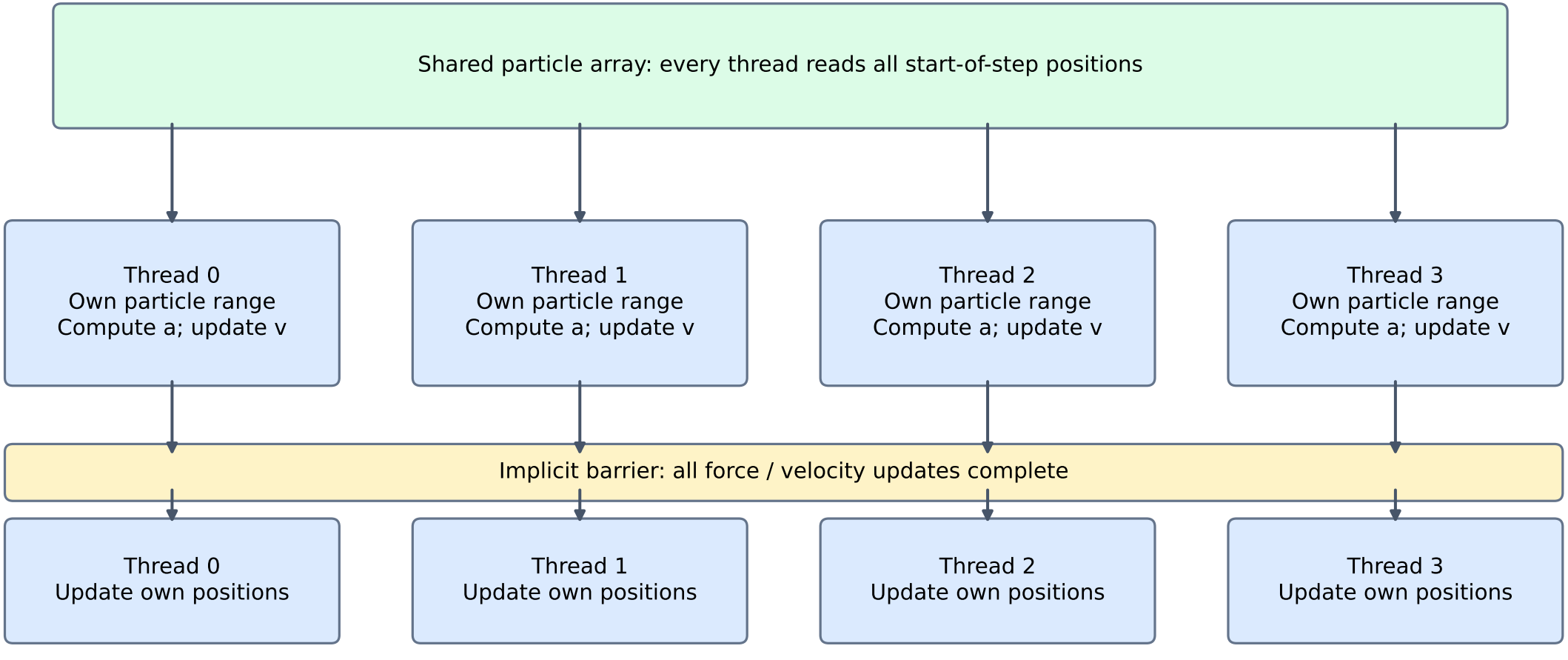
OpenMP: shared positions, disjoint updates

Example with four threads and N divisible by four; static scheduling distributes outer-loop particles



A second implicit barrier completes the position loop before the next timestep. No particle is updated by two threads.